

Bread components derived from yeast-fermented wheat and rye breads increase colitis and mortality in a mouse model of colitis

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Supplementary material

Table S1: Amount of Oligosaccharides in the different breads.

Bread	Oligosaccharides DP≤8 [g/100g]	Oligosaccharides DP>8 [g/100g]	Total Oligosaccharides [g/100g]
W SF Y	3.6	1.6	5.3
W SF S	3.3	1.4	4.6
W SF SB	4.4	2.0	6.4
W WG Y	4.2	1.6	5.8
Sp SF Y	1.8	1.4	3.2
R SF Y	5.8	2.2	8.0

Abbreviations: W, Wheat; Sp, Spelt; R, Rye; SF, Superfine flour; WG, Wholegrain flour; Y, Yeast fermented; S, Sourdough fermented; DP, Degree of Polymerization.

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Table S2. Average food intake and liver/body weight ratio, body weight changes in Casp8^{ΔIEC} and Casp8^{fl} mice.

	Genotype	GF	5G	WSFY	WSFS	WSFSB	WWGY	SpSFY	RSFY
Food intake/ mouse [g/day]	Casp8 ^{ΔIEC}	2.6	2.5	2.5	2.4	2.7	2.8	2.5	2.6
		±0.26	±0.25	±0.23	±0.05	±0.12	±0.14	±0.21	±0.15
	Casp8 ^{fl}	2.5	2.8	2.4	2.5	2.7	2.9	2.7	2.6
		±0.00	±0.14	±0.05	±0.15	±0.10	±0.15	±0.17	±0.12
Liver/Body weight ratio [g]	Casp8 ^{ΔIEC}	0.05	0.04	0.04	0.05	0.05	0.04	0.05	0.05
		±0.00	±0.00	±0.4	±0.01	±0.00	±0.00	±0.01	±0.01
	Casp8 ^{fl}	0.04	0.03	0.04	0.04	0.04	0.04	0.04	0.04
		±0.00	±0.01	±0.00	±0.01	±0.00	±0.00	±0.00	±0.01
Body weight changes [%]	Casp8 ^{ΔIEC}	20.3	8.6	2.9	2.4	7.5	2.4	2.4	-3.7
		±6.45	±3.65	±5.80*	±3.99*	±2.50	±5.49*	±4.18*	±5.23***
	Casp8 ^{fl}	13.8	12.1	10.5	12.0	16.6	14.4	11.0	10.2
		±2.99	±1.68	±1.75	±1.84	±2.48	±2.93	±2.54	±1.56

Mice were fed a gluten-free diet (GF), GF with 5% gluten added (5G) or a diet with 50% bread. Values are means ± SEM, statistics by two-factor ANOVA. Group differences were assessed by the Tukey's post hoc test. Only differences compared to the GF diet were found and indicated by asterisks (*P<0.05; ***P<0.001). No genotype-specific differences were found. Abbreviations: W, Wheat; Sp, Spelt; R, Rye; SF, Superfine flour; WG, Wholegrain flour; Y, Yeast fermented; S, Sourdough fermented.

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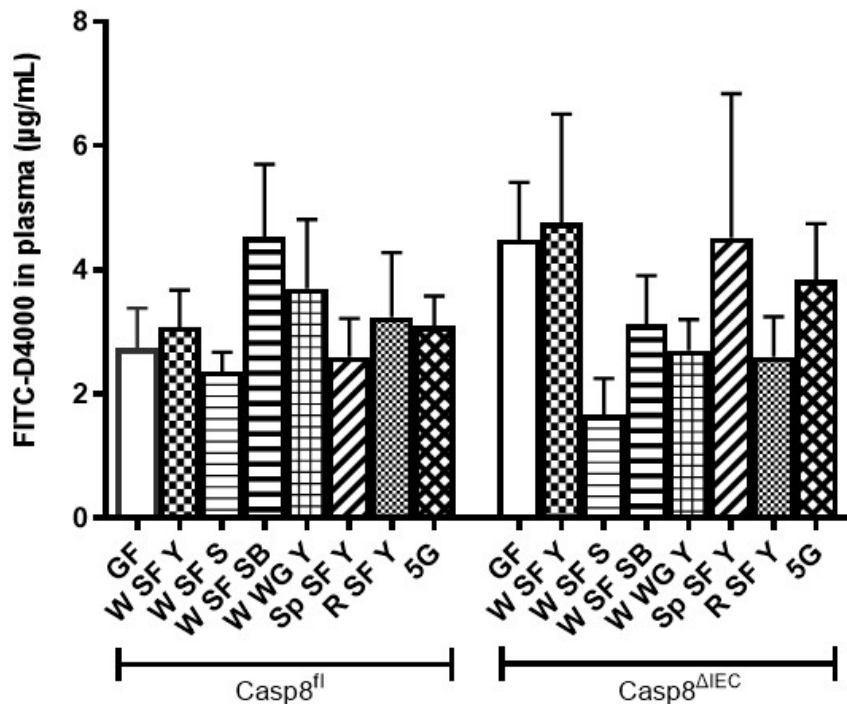


Figure S1: Small intestinal barrier analyzed by FITC-Dextran 4000 in plasma. Abbreviations see Table S1.